

[0:00 – 0:45] Slide 1: Introduction & Welcome

Speaker:

"Welcome everyone, and thank you for joining today's briefing on Denial-of-Service and Distributed Denial-of-Service attacks.

In today's interconnected digital economy, availability is the backbone of business trust. Yet, availability remains one of the most frequently targeted pillars of cybersecurity. Every minute an online service is unreachable translates directly into lost revenue, brand degradation, and operational chaos.

Over the next five minutes, we will break down the fundamental mechanics of DoS versus DDoS attacks, examine the three core attack vectors, look at real-world scale metrics, and review modern multi-layer defense architectures."

[0:45 – 1:30] Slide 2: DoS vs. DDoS Mechanics

Speaker:

"Let's begin by distinguishing Denial-of-Service from Distributed Denial-of-Service.

A traditional **Denial-of-Service (DoS)** attack originates from a single source computer targeting a single host. The attacker sends malicious traffic or exploits a flaw to exhaust CPU or memory. Because it comes from one IP address, defense is relatively straightforward: block the IP.

However, a **Distributed Denial-of-Service (DDoS)** attack amplifies this threat exponentially. The attacker controls a botnet—thousands or millions of compromised devices, such as infected IoT cameras, routers, or servers. These hosts flood the target concurrently. Blocking

individual IP addresses becomes nearly impossible without dropping legitimate user traffic."

[1:30 – 2:30] Slide 3: The 3 Primary Attack Vectors

Speaker:

"DDoS attacks are not monolithic. They fall into three primary categories based on where they strike in the network stack.

First, **Volumetric Attacks**. The objective here is pure bandwidth saturation. Attackers flood network interfaces with massive streams of UDP or ICMP traffic, filling pipe capacity so legitimate packets cannot reach the host.

Second, **Protocol Attacks**. These target network infrastructure and state tables, such as firewalls and load balancers. By exploiting protocols through SYN floods or CLDAP amplification, attackers exhaust server resources without needing massive bandwidth.

Third, **Application-Layer Attacks**. Operating at Layer 7, these mimic legitimate user behavior with HTTP GET or POST requests. They target expensive operations like database queries or login endpoints, starving server CPU with low network volume."

[2:30 – 3:30] Slide 4: Modern Threat Landscape & Scale

Speaker:

"To appreciate why legacy firewalls no longer suffice, look at current threat statistics.

The scale of modern attacks has reached hyper-volumetric levels. Industry intelligence reports peak DDoS assaults surpassing **31.4 Terabits per second**—driven by automated IoT botnets.

Globally, autonomous systems mitigate over **5,300 DDoS attacks every single hour**.

Critically, over **90% of modern network attacks end in under 10 minutes**. Attackers use short, hyper-dense bursts to disrupt service before security teams can manually respond. This makes manual intervention obsolete; mitigation must be instantaneous and automated."

[3:30 – 4:30] Slide 5: Multi-Layer Defense Architecture

Speaker:

"Defending against modern DDoS requires a multi-layered defense pipeline operating upstream.

First, **Anycast DNS Routing** distributes incoming traffic across a globally decentralized network of edge locations, absorbing volumetric spikes near the traffic source.

Second, **Scrubbing Centers** analyze traffic using BGP rerouting, filtering malicious packets while passing clean traffic back to origin servers.

Third, at the host level, **eBPF and XDP kernel filters** drop malicious network packets directly at line rate before they reach the application OS stack.

Finally, a **Web Application Firewall (WAF)** enforces Layer-7 rate limiting and challenge rules to stop stealthy application floods."

[4:30 – 5:00] Slide 6: Key Takeaways & Strategic Posture

Speaker:

"To summarize our key takeaways for securing enterprise infrastructure:

1. **Always-On Mitigation:** With attacks peaking in seconds, reactive security fails. Mitigation must be continuous and automated.
2. **Multi-Vector Defense:** Protection must filter Layer 3 through Layer 7 simultaneously to catch blended campaigns.
3. **Upstream Capacity:** You cannot block a 1-Terabit flood at your local firewall. Offload traffic to specialized cloud mitigation networks."

[5:00 – 5:30] Slide 7: References & Q&A

Speaker:

"The data presented in this briefing is sourced from global threat reports published by Cloudflare, SentinelOne, Radware, and StormWall Network Analysis.

Thank you for your time today. We will now open the floor for any questions."